

MISAWA AB, Japan

WMO: 475800

Lat: 40.705N

Lon: 141.372E

Elev: 39

StdP: 100.85

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 44402

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-6.2	-12.8	1.2	-4.3	-11.8	1.4	-3.8	13.5	-0.8	12.6	-1.1	4.7	280	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	30.0	23.7	28.1	22.7	26.8	22.2	24.8	28.4	23.9	26.8	23.1	25.8	5.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.8	27.0	23.0	17.8	25.9	22.1	16.8	24.9	75.6	28.5	71.8	26.3	68.7	25.8	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.7	9.4	-10.1	33.0	1.6	1.6	-11.3	34.2	-12.3	35.1	-13.2	36.0	-14.4	37.1		
(5)			-10.7	26.2	1.5	1.3	-11.8	27.1	-12.7	27.9	-13.5	28.6	-14.6	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-1.7	-1.0	2.4	8.3	13.2	16.2	20.5	22.4	19.2	13.2	6.9	1.0		
	DBStd	8.84	2.60	2.91	3.37	3.56	3.40	3.05	3.51	2.97	2.96	3.01	3.75	3.12		
	HDD10.0	1381	362	309	236	75	10	0	0	0	0	6	104	278		
	HDD18.3	3282	621	543	493	303	164	77	18	3	22	160	342	537		
	CDD10.0	1424	0	0	1	23	108	187	326	384	277	107	13	0		
	CDD18.3	282	0	0	0	1	3	14	85	128	50	2	0	0		
	CDH23.3	1493	0	0	0	4	28	71	482	710	196	3	0	0		
	CDH26.7	348	0	0	0	1	4	11	114	184	34	0	0	0		
(14) Wind	WSAvg	4.3	4.9	4.8	5.1	5.0	4.2	3.5	3.4	3.5	3.7	4.0	4.4	4.7		
(15) Precipitation	PrecAvg	1173	67	54	67	70	90	102	142	143	165	110	85	72		
	PrecMax	1483	131	153	116	148	142	203	277	328	322	217	163	189		
	PrecMin	886	27	20	21	15	32	32	24	26	54	27	28	36		
	PrecStd	159	27	28	27	33	34	41	64	83	67	55	37	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	10.2	16.2	23.0	26.2	28.1	31.9	32.9	30.1	23.2	18.9	12.2		
		MCWB	4.0	6.3	9.1	12.8	15.6	19.4	24.7	24.7	23.6	17.2	13.9	8.8		
	2%	DB	5.0	7.0	12.8	19.1	23.2	25.1	29.9	30.2	27.2	21.1	16.2	9.8		
		MCWB	2.4	3.9	7.5	11.0	15.0	18.7	23.5	24.2	22.1	16.5	12.1	7.1		
	5%	DB	3.8	5.2	10.1	16.8	21.2	23.1	27.9	28.9	25.2	19.9	14.8	8.0		
		MCWB	1.6	2.2	5.8	9.8	14.2	18.0	22.8	23.6	20.8	15.7	11.2	5.6		
	10%	DB	2.2	3.9	8.1	14.2	19.1	21.2	26.1	27.0	24.0	18.2	13.0	6.0		
		MCWB	0.2	1.3	4.4	8.9	13.1	17.2	22.1	22.9	20.1	14.6	9.9	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	6.9	10.2	14.7	18.1	21.8	25.9	26.2	24.5	19.5	15.1	9.9		
		MCDB	6.1	9.6	14.2	19.7	22.7	25.3	29.8	30.1	28.3	21.6	17.0	11.7		
	2%	WB	2.8	4.1	7.8	12.5	16.4	20.3	24.5	25.2	23.1	17.6	12.9	7.4		
		MCDB	4.4	6.4	11.4	16.9	21.2	23.5	28.1	28.9	26.2	19.8	15.7	9.2		
	5%	WB	1.5	2.6	6.2	10.8	15.2	18.9	23.5	24.3	22.1	16.5	11.5	5.5		
		MCDB	3.1	4.8	9.6	15.0	19.6	22.1	26.7	27.3	24.7	18.8	14.1	7.4		
	10%	WB	0.5	1.3	4.7	9.7	14.0	17.6	22.6	23.5	21.0	15.5	10.3	3.7		
		MCDB	1.9	3.4	7.4	13.4	17.9	20.6	25.6	26.2	23.0	17.7	12.6	5.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.1	7.4	9.1	8.5	6.8	6.1	6.0	7.2	8.4	7.8	6.1		
		MCDBR	7.5	8.6	10.8	12.7	12.4	10.3	9.2	8.8	8.4	9.4	10.4	8.8		
	5% WB	MCWBR	6.0	6.7	7.5	7.2	6.4	5.3	4.7	4.1	4.9	6.1	7.3	7.0		
		MCWBR	7.0	8.4	10.4	11.0	10.5	8.7	8.2	7.5	7.3	8.0	9.6	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.364	0.406	0.476	0.524	0.539	0.520	0.522	0.493	0.437	0.412	0.394	0.374		
	taud		2.078	1.959	1.870	1.835	1.879	2.011	2.079	2.170	2.281	2.225	2.166	2.109		
	Ebn at Noon		752	773	762	757	758	773	767	778	799	769	717	702		
	Edh at Noon		113	148	181	201	197	173	160	143	120	113	104	102		
(44) All-Sky Solar Radiation	RadAvg		1.53	2.26	3.36	4.46	4.92	4.83	4.29	4.14	3.60	2.79	1.80	1.30		
	RadStd		0.07	0.19	0.24	0.42	0.51	0.37	0.48	0.49	0.29	0.24	0.14	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	-0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page